

Lab 30: NumPy Random Module for Simulation

Aim

To generate random numbers using **NumPy's random module** for simulation and sampling tasks.

Algorithm

1. Import the NumPy library.
2. Create a seeded random number generator.
3. Generate uniform random numbers.
4. Generate random integers.
5. Generate samples from a normal distribution.
6. Perform a simple Monte Carlo simulation to estimate the value of π .
7. Display the generated values and estimated π .

Program

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```
import numpy as np
```

```
# Create seeded random number generator
```

```
rng = np.random.default_rng(seed=42)
```

```
# Generate random values
```

```
print("Uniform random 2x3:\n", rng.random((2, 3)))
```

```
print("Random integers 0-10:", rng.integers(0, 10, size=5))
```

```
print("Gaussian samples:", rng.normal(0, 1, size=5))
```

```
# Monte Carlo simulation to estimate pi
```

```
n = 100000
```

```
points = rng.random((n, 2))
```

```
inside = np.sum(  
    points[:, 0] ** 2 + points[:, 1] ** 2 <= 1  
)
```

```
pi_estimate = 4 * inside / n
```

```
print("Estimated pi:", pi_estimate)
```

Sample Output

Uniform random 2x3:

```
[[0.774 0.439 0.858]
```

```
[0.697 0.094 0.976]]
```

Random integers 0-10: [0 7 6 4 4]

Gaussian samples: [0.305 -0.588 1.298 -0.089 0.523]

Estimated pi: 3.14156

Result

Thus, NumPy's random module was successfully used to generate reproducible random samples and perform a Monte Carlo simulation to estimate π .